Elementary Problems
Divisibility

Version 1.0

2-9-2018

Basic Work

1. Prove that $8 \mid n^2 - 1$ for any odd integer n.
2. Find $(1234, 5678)$.
3. Find $(a + b, a - b)$ where a and b are relatively prime integers.
4. Find $(n^{120} - 1, n^{35} - 1)$ where n is a positive integer greater than 1.
5. (IMO 1959 Problem 1) Prove that the fraction $\dfrac{21n + 4}{14n + 3}$ is irreducible for every natural number n.
6. Let a, b, c, d be integers such that $(a, c) = (a, d) = (b, c) = 1$. Prove that $(ac, ab+cd) = 1$.
7. The sequence $(a_n)_{n\geqslant 1}$ is defined by $a_1 = a_2 = 1$ and $a_{n+2} = 4a_{n+1} + a_n$ for $n \geqslant 1$. Prove that any two consecutive terms of the sequence are relatively prime.
8. The sequence $(a_n)_{n\geqslant 1}$ is defined by $a_n = n^2 + 20$ for $n \geqslant 1$. Prove that the greatest common divisor of any two consecutive terms of the sequence divides 81.
9. Let n be an odd positive integer. Let $a_1, a_2, \ldots, a_n$ be a permutation of $1, 2, \ldots, n$. Prove that
$$(a_1 - 1)(a_2 - 2) \cdots (a_n - n)$$
is even.
10. Find all integers $n \neq -1$ such that $n + 1 \mid n^2 + 1$.
11. Find all positive integers $n \neq 4$ such that $n - 4 \mid 2n + 6$.
12. Find all positive integers n such that $3n - 1 \mid 2n + 2$.
13. Let m and n be integers such that $17 \mid 2m + 3n$. Prove that $17 \mid 9m + 5n$.
14. Prove that $\sqrt{2}$ is an irrational number.
15. Prove that $\sqrt{2} + \sqrt{3}$ is an irrational number.
16. Let n be a positive integer. Prove that there exist n consecutive composite numbers.
17. Let n be a positive integer. Prove that the product of any n consecutive positive integers is divisible by $n!$.
18. Let $m, n > 1$ be integers. If $m^n + 1$ is a prime number, prove that n is a power of 2.

19. (Finnish National High School Mathematics Competition 1999) Determine how many primes are there in the sequence

$$101, 10101, 1010101, \ldots.$$

20. (MEMO 2008 Problem 4) Determine all integers k such that for all n the numbers $4n+1$ and $kn+1$ have no common divisor.

21. Let m and n be odd positive integers such that $m^n n^m$ is a perfect square. Prove that mn is a perfect square.

22. Find all positive integers m and n such that $\frac{m}{n} + \frac{n}{m}$ is a positive integer.

23. Find all positive integers m and n such that $4m^2 - n^2 \mid 2m^2 - n^2$.

24. (IMO shortlist 1990) Find all natural numbers n for which every natural number whose decimal representation has $n-1$ digits 1 and one digit 7 is prime.

25. Find all nonempty subsets S of the set of integers such that for any $x, y \in S$ (not necessarily distinct), we have $x - y \in S$.

26. Prove or disprove the following.

    (a) If a, b, c are positive integers such that $a \mid bc$, then $a \mid b$ or $a \mid c$.

    (b) If m and n are positive integers such that $m^2 \mid n^2$, then $m \mid n$.

    (c) If m and n are positive integers such that $mn \mid m+n$, then $m = n$.

    (d) If m and n are even positive integers such that $m+n \mid mn$, then $m = n$.

    (e) Let $P(x)$ and $Q(x)$ be polynomials with integer coefficients such that $P(x)$ does not divide $Q(x)$. Then there are finitely many positive integers n such that $P(n) \mid Q(n)$.

    (f) If m and n are positive integers, then $(m^2, n^2) = (m, n)^2$.

    (g) If a, b, c are positive integers such that $(a, b, c) = 1$, then at least two of a, b, c are relatively prime.

    (h) If m and n are positive integers whose least common multiple is mn, then $(m, n) = 1$.

Prime Number

27. Prove that there are infinitely many prime numbers.

28. Let p be a prime number and n an integer. If $p \mid n^2$, prove that $p^2 \mid n^2$.

29. Let p be a prime number larger than 5. Prove that $240 \mid p^4 - 1$.

30. (Iran MO 2nd Round 1996 Problem 2) Let a, b, c, d be positive integers such that $ab = cd$. Prove that $a + b + c + d$ is a composite number.

31. (IMO 1979 Problem 1) Let p and q be natural numbers such that

$$\frac{p}{q} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots - \frac{1}{1318} + \frac{1}{1319}.$$

Prove that p is divisible by 1979.

32. (IMO 1970 Problem 4) Find the set of all positive integers n with the property that the set $\{n, n+1, n+2, n+3, n+4, n+5\}$ can be partitioned into two sets such that the product of the numbers in one set equals the product of the numbers in the other set.

33. (Canada MO 1992 Problem 1) Prove that the product of the first n natural numbers is divisible by the sum of the first n natural numbers if and only if $n+1$ is not an odd prime.

34. (IMO shortlist 1967) Let k, m, n be natural numbers such that $m + k + 1$ is a prime greater than $n + 1$. Let $c_s = s(s+1)$. Prove that

$$(c_{m+1} - c_k)(c_{m+2} - c_k) \cdots (c_{m+n} - c_k)$$

is divisible by the product $c_1 c_2 \cdots c_n$.

35. Find all prime numbers p and q such that $p^2 \mid q^3 + 1$ and $q^2 \mid p^3 + 1$.

36. Let p be a prime number such that $p = m^2 + n^2$ for some positive integers m and n. If $p \mid (m+n)^3 + 8$, find all possible values of p.

<u>Size</u>

37. Find all positive integers n such that $3n - 1 \mid 2n + 2$.

38. Find all positive integers n such that $n^2 + 1 \mid 2n + 1$.

39. Find all positive integers n such that $2n^2 + 3 \mid n^3 + 3n + 6$.

40. Find all positive integers n such that $n^2 + 2 \mid 3n - 12$.

41. Find all positive integers m and n such that $mn \mid m + n$.

42. Find all positive integers m and n such that $m \mid n + 1$ and $n \mid m + 1$.

43. Find all positive integers a, b, c with $abc > 1$ such that $abc - 1 \mid ab^2$.

44. Find all positive integers m and n such that $m \mid 2n + 1$ and $n \mid 3m + 2$.

45. Find all positive integers m and n such that $m^2 - n^2 \mid m + 4n$.

46. (IMO 1992 Problem 1) Find all integers a, b, c with $1 < a < b < c$ such that

$$(a-1)(b-1)(c-1)$$

is a divisor of $abc - 1$.

47. Find all positive integers a, b, c such that $a \mid bc - 1$, $b \mid ca - 1$ and $c \mid ab - 1$.

48. (All-Russian MO 1964) Let a, b, n be natural numbers such that for every natural number $k \neq b$, $b - k$ divides $a - k^n$. Prove that $a = b^n$.

49. (Spain MO 2016 Problem 4) Let m be a positive integer and a and b be distinct positive integers strictly greater than m^2 and strictly less than $m^2 + m$. Find all integers d such that $m^2 < d < m^2 + m$ and d divides ab.

50. (Czech-Polish-Slovak Match 2002 Problem 4) An integer $n > 1$ and a prime p are such that n divides $p - 1$, and p divides $n^3 - 1$. Prove that $4p - 3$ is a perfect square.

51. Find all positive integers $m, n > 1$ such that $mn - 1 \mid m^2 - n$.

Infinite Descent and Vieta Jumping

52. Find all integer solutions to $a^3 + 2b^3 + 4c^3 = 0$.

53. Do there exist positive integers a, b, c such that all of $a + 2b$, $b + 2c$, $c + 2a$ are powers of 2?

54. Let k be a positive integer. There are $2k + 1$ positive integers. If we remove any of these positive integers, the remaining $2k$ numbers can be partitioned into two groups of k integers with the same sum. Prove that all the positive integers are equal.

55. (CWMO 2010 Problem 8) Find all integers k such that there exist positive integers a and b satisfying $\dfrac{b+1}{a} + \dfrac{a+1}{b} = k$.

56. Suppose m and n are positive integers such that $mn \mid m^2 + n^2 + 6$. Find all possible values of $\dfrac{m^2 + n^2 + 6}{mn}$.

57. (IMO 1988 Problem 6) Let a and b be positive integers such that $ab + 1$ divides $a^2 + b^2$. Show that

$$\frac{a^2 + b^2}{ab + 1}$$

is the square of an integer.

58. (IMO shortlist 1992) Prove that for any positive integer m there exist an infinite number of pairs of integers (x, y) such that

(i) x and y are relatively prime;

(ii) y divides $x^2 + m$;

(iii) x divides $y^2 + m$;

(iv) $x + y \leqslant m + 1$.

Record

v1.0: problems 1–58